ARIMA Forecasting and Event Detection: Linear Time Series Models

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Predictive Modeling

The model defines the expected behavior of the series; deviations are identified where the forecast and observed value diverge beyond a threshold h.

$$E_t = \begin{cases} 1, & |X_t - \hat{X}_t| > h \\ 0, & \text{otherwise} \end{cases}$$

The residual $R_t = X_t - \hat{X}_t$ plays a **dual role**: it evaluates the quality of the forecast (how well the model fits the series) and serves as the basis for deviation detection (when $|R_t| > h$, an event is flagged).

Observed Value

X_t — actual data at time t

Forecast Value

$\hat{X}_t$ — normality reference generated by the model

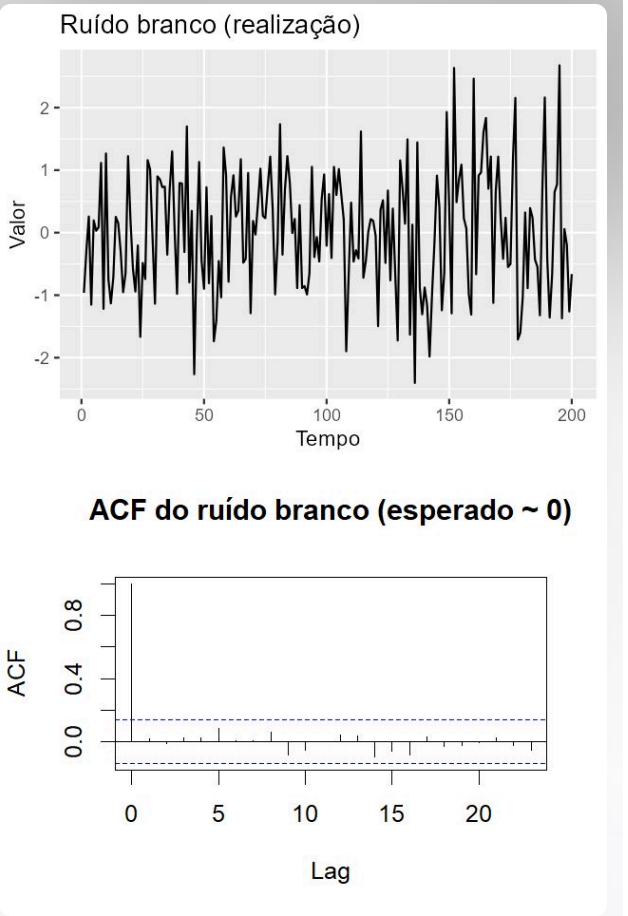
Threshold

h — defines what is considered a significant deviation

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INPUT: X (series of length T), model M, h (threshold)
PRE-CONDITION: model M fitted on X[1:W]; W defined before the loop begins

STEP 1 — Initial training period
  W ← minimum training window size
  Fit M on X[1 : W]
STEP 2 — One-step-ahead forecast generation
  FOR t from W+1 to T:
    X̂_t ← M.predict(X[1 : t-1])   // only information prior to t
STEP 3 — Residual calculation
  FOR t from W+1 to T:
    R_t ← X_t - X̂_t
STEP 4 — Threshold application
  FOR t from W+1 to T:
    IF |R_t| > h:
      E_t ← 1   // deviation detected
    ELSE:
      E_t ← 0   // normal behavior

OUTPUT: E (binary deviation indicator per time step)
POST-CONDITION: E_t = 1 identifies time steps where the
  model does not explain the observed behavior
```



Fundamentals of White Noise

Zero Mean

The expectation of white noise is always zero, representing the absence of systematic bias.

$$\mathbb{E}[\varepsilon_t] = 0$$

Constant Variance

The variance remains constant over time, ensuring homoscedasticity.

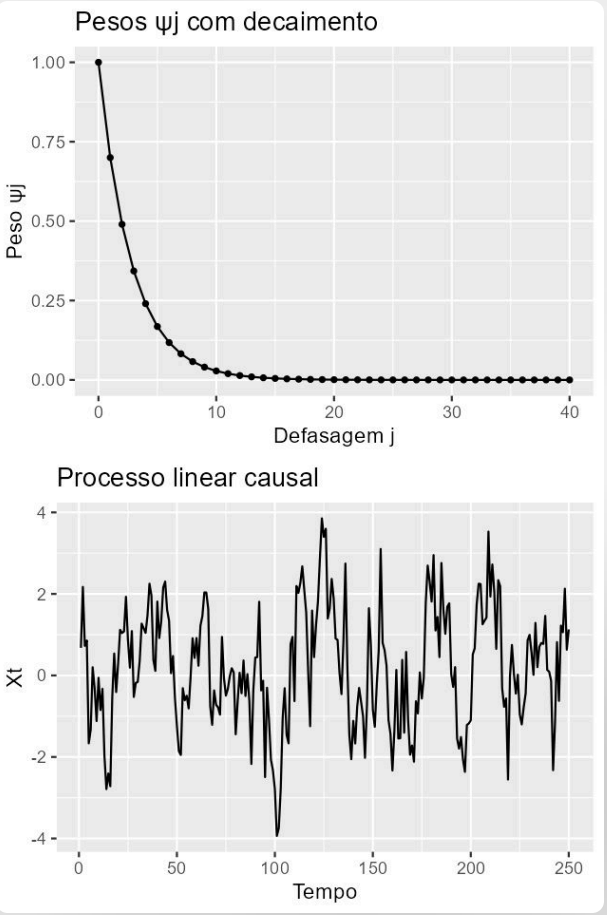
$$\mathrm{Var}(\varepsilon_t) = \sigma^2$$

Uncorrelated

Shocks at different time points are independent of each other.

$$\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-h}) = 0$$

These three properties define white noise as the fundamental unpredictable component in time series models.



Causal Linear Process

A causal linear process expresses the current value of the series as a function of past and present random shocks:

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$$

Where μ represents the mean of the process, ε_t are the random shocks, and ψ_j are the weights that determine how each lagged shock influences the present.

Causality ensures that the present depends only on the past, not the future. The weights ψ_j typically decrease with j, reflecting that older shocks have less impact.

Current Shock

Immediate impact ψ_0

Recent Shocks

Moderate influence

Older Shocks

Diminishing effect

Lag Operator

The lag operator B is an algebraic tool that shifts the series in time. Its fundamental definition is $BX_t = X_{t-1}$, and successive applications yield $B^k X_t = X_{t-k}$.

Autoregressive Polynomial

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$$

Represents the dependence structure of past values of the series.

Moving Average Polynomial

$$\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$$

Captures the propagation of random shocks over time.

These polynomials transform temporal models into compact algebraic expressions, simplifying mathematical analysis and the formulation of stationarity conditions.

AR(1): First Model of Temporal Dependence

The Model

$$X_t - \mu = \phi\,(X_{t-1} - \mu) + \varepsilon_t$$

$$\hat{X}_t = \mu + \phi\,(X_{t-1} - \mu)$$

Temporal Memory

The current value carries information from the past.

Parameter φ

Controls the intensity of dependence. The larger |φ|, the stronger the memory.

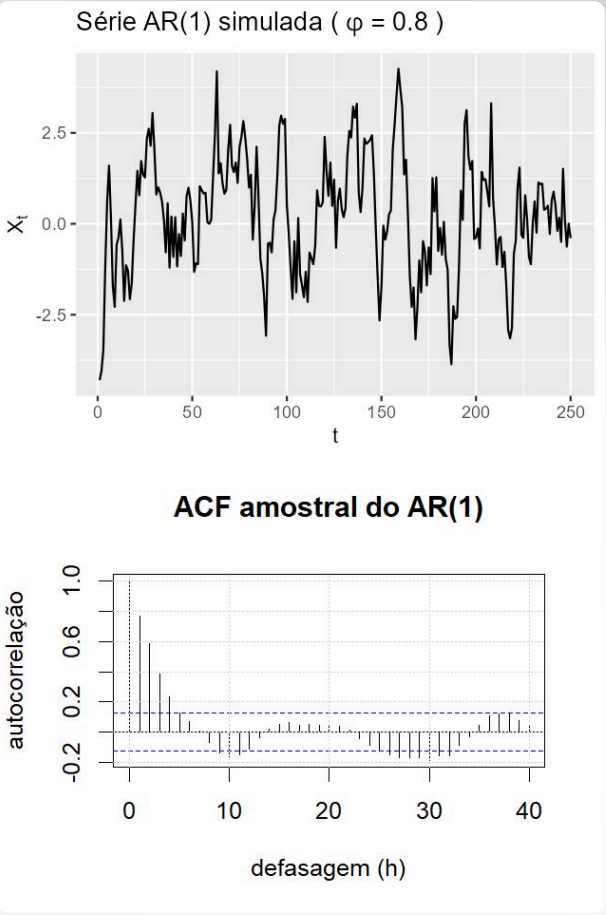
Residual ε_t

Random shock observed AFTER x_t is known: ε_t = x_t − x̂_t. Does not enter the operational forecast formula.

```
INPUT: series X = <x1, ..., xn>
PRE-CONDITION: stationary or centered series; n >= 2

STEP 1 — Estimate the mean mu of the series
STEP 2 — Center the series:
  y_t <- x_t - mu
STEP 3 — Estimate phi:
  phi <- sum_{t=2..n}(y_t * y_{t-1}) /
         sum_{t=2..n}(y_{t-1}^2)
STEP 4 — For t = 2 to n:
  yhat_t    <- phi * y_{t-1}
  xhat_t    <- mu + yhat_t
  epsilon_t <- x_t - xhat_t
STEP 5 — Diagnose residuals epsilon_t

OUTPUT: phi, fitted values xhat_t and residuals epsilon_t
POST-CONDITION: residuals must behave as white noise
```



AR(1): Conditions and Properties

Stationarity

The AR(1) process is stationary when $|\phi| < 1$. In this case, the effects of past shocks diminish over time and the statistical properties stabilize.

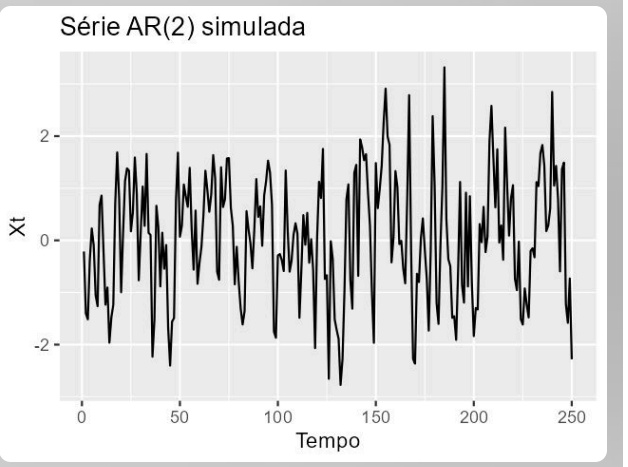
$$|\phi| < 1$$

Autocorrelation

The autocorrelation of AR(1) decays exponentially with the lag. This "signature" allows the model to be identified from observed data.

$$\rho(h) = \phi^h$$

AR(1) is the first concrete example of how temporal dependence can be formalized and estimated. It illustrates the principles underlying more complex models — higher-order AR, MA, ARMA, and ARIMA — which will be explored in depth in presentation 04.



Autoregressive Model AR(p)

In the autoregressive model, the series is explained by its own past values plus a white noise term representing unpredictable innovation:

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$$

Expanded Form

Explicit linear combination of p lagged values

Compact Form

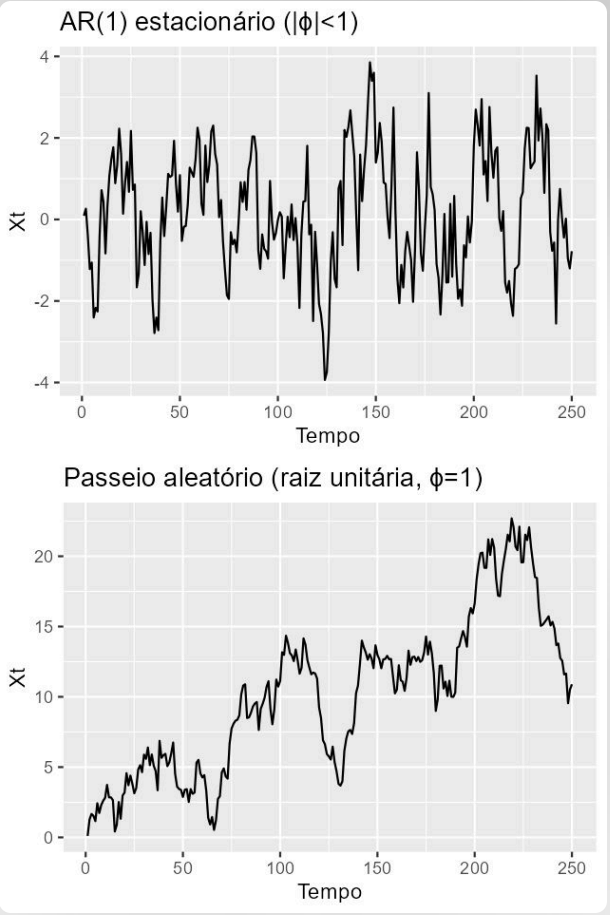
Representation using the operator:

$\phi(B)X_t = c + \varepsilon_t$

Process Mean

Determined by the constant c and the autoregressive coefficients ϕ_i

The constant c defines the mean level of the series, while the coefficients ϕ_i determine the intensity of temporal dependence.



Stationarity of AR(p)

The stationarity of an AR(p) process depends fundamentally on the autoregressive coefficients. The condition is expressed through the roots of the characteristic polynomial:

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$$

For stationarity, all roots must satisfy $|z| > 1$, that is, they must lie outside the unit circle in the complex plane.

Special Case: AR(1)

For the model $X_t = c + \phi X_{t-1} + \varepsilon_t$, the condition simplifies to $|\phi| < 1$.

Properties of AR(1)

As a special case of AR(p) with $p = 1$, the AR(1) directly illustrates the general properties of the model:

$$X_t = c + \phi X_{t-1} + \varepsilon_t$$

Mean

Special case of $\mu = (I - \Phi)^{-1} c$: $\mu = \frac{c}{1-\phi}$

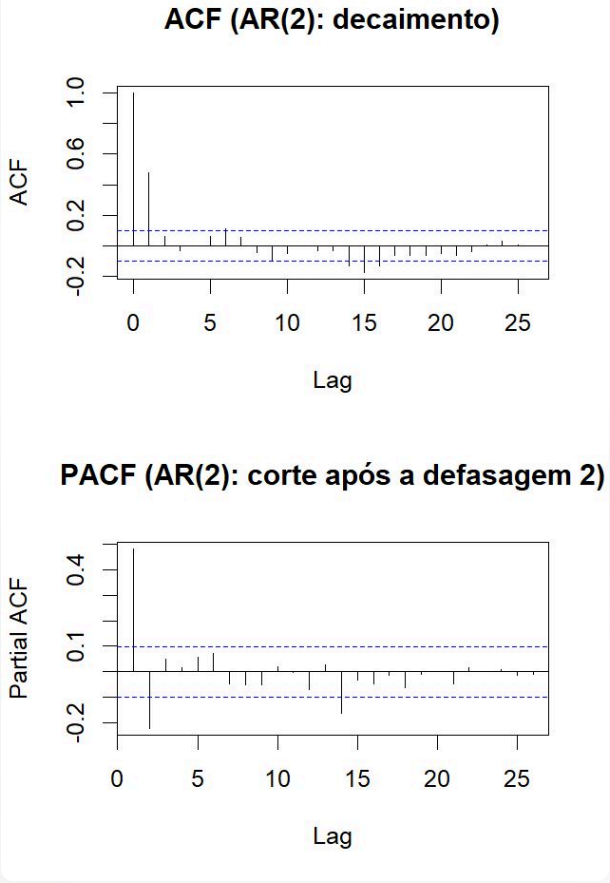
Variance

Special case of the Yule-Walker equation:
$\operatorname{Var}(X_t) = \frac{\sigma^2}{1-\phi^2}$

Autocorrelation

Exponential decay of the general ACF:
$\rho(h) = \phi^h$

The exponential decay structure of the AR(1) ACF anticipates the behavior of multi-step forecasting: $\phi^h \to 0$ when $|\phi| < 1$.



Partial Autocorrelation (PACF)

The partial autocorrelation measures the direct dependence between X_t and X_{t-h}, removing the effect of all intermediate lags.

Mathematically, $\alpha(h)$ is the coefficient of X_{t-h} in the regression:

$$X_t = \beta_1 X_{t-1} + \cdots + \beta_h X_{t-h} + u_t$$

Fundamental Property

For an AR(p) process, the PACF satisfies $\alpha(h) = 0$ for all $h > p$

Order Identification

The sharp cutoff of the PACF directly identifies the order p of the autoregressive model

The PACF is an essential tool for model identification, complementing the analysis of the ACF.

Moving Average Model MA(q)

In the moving average model, the series is explained by present and past random shocks, without direct dependence on past values of the series itself:

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Compact Form

Using the lag operator: $X_t = \mu + \theta(B)\varepsilon_t$

Structure

Linear combination of $q+1$ consecutive random shocks

Finite Memory

Only the last q shocks influence the present

Unlike AR, there is no regression on past values of the series. The temporal behavior is determined exclusively by the accumulation of random shocks.

ARMA(p,q) Model

The ARMA model combines autoregressive and moving average components, representing the most general linear form for stationary time series:

$$X_t = c + \sum_{i=1}^{p} \phi_i X_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

AR Part

Captures temporal persistence through p lagged values

MA Part

Models shock propagation through q error terms

Compact Form

Expressed as $\phi(B)X_t = c + \theta(B)\varepsilon_t$

ARMA is the foundation of classical stationary time series models, offering flexibility to capture various temporal dependence patterns with parameter parsimony.

Stationarity and Invertibility Conditions

Two conditions ensure that the ARMA(p,q) is well-defined and interpretable.

Stationarity (AR part)

Condition: $\phi(z) = 0 \Rightarrow |z| > 1$

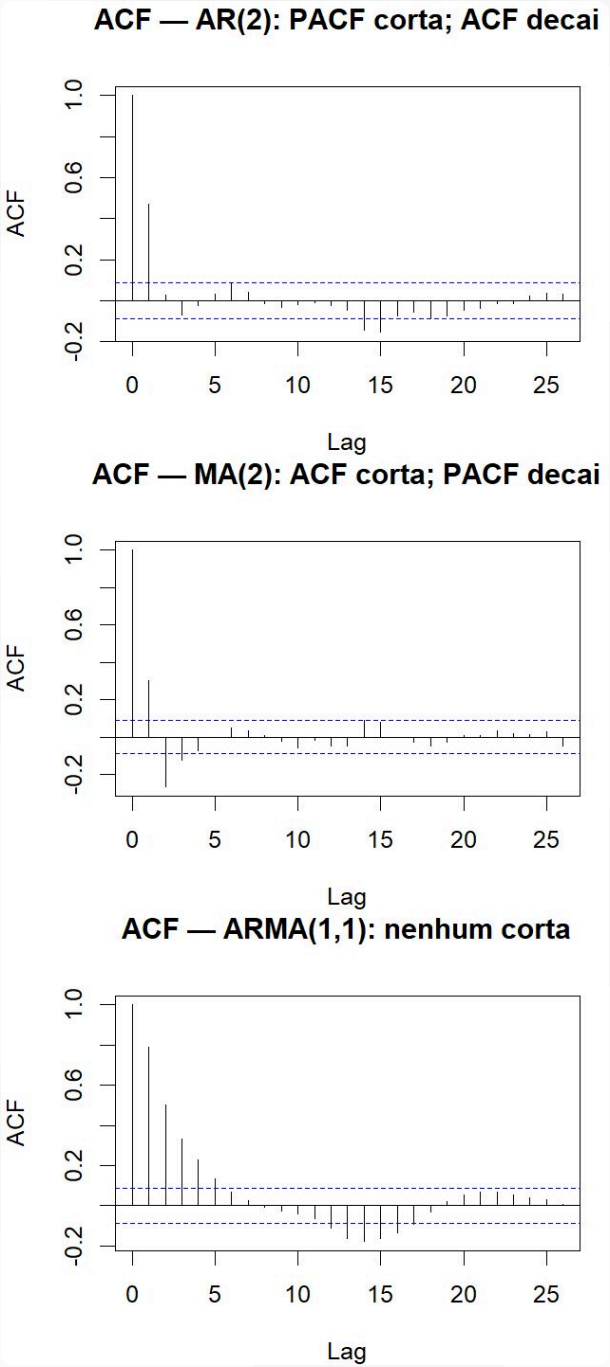
Roots of the AR polynomial outside the unit circle. Ensures the series does not explode and that its statistical properties are stable over time.

Invertibility (MA part)

Condition: $\theta(z) = 0 \Rightarrow |z| > 1$

Roots of the MA polynomial outside the unit circle. Ensures uniqueness of the representation and that past shocks can be recovered.

Under these conditions, the ARMA admits equivalent representations as MA(∞) and AR(∞), revealing the structural duality of linear models.



ACF and PACF in ARMA

AR(p)

PACF: Cuts off after lag p

ACF: Decays gradually

MA(q)

ACF: Cuts off after lag q

PACF: Decays gradually

ARMA(p,q)

ACF: Decays without abrupt cutoff

PACF: Decays without abrupt cutoff

In ARMA, the absence of abrupt cutoffs requires information criteria to identify the orders.

Box-Jenkins Method

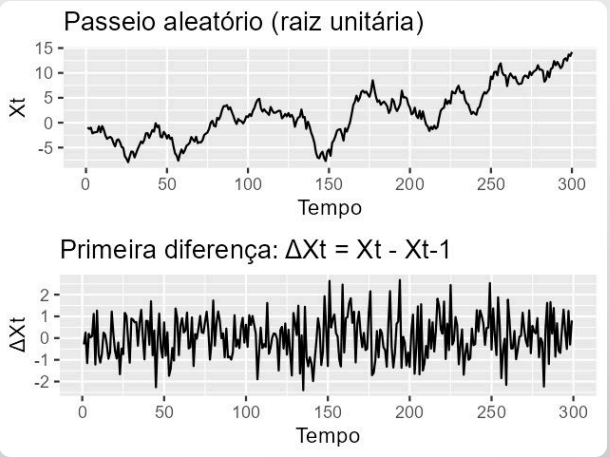
Iterative four-step procedure for building an ARMA(p,q) model ready for forecasting.

$$\phi(B)X_t = c + \theta(B)\varepsilon_t$$

AIC and BIC criteria penalize models with more parameters, favoring parsimony.

```
INPUT: Time series {X_t}, maximum orders p_max, q_max
PRECONDITION: Stationary series (or previously differenced)

STEP 1 — Identification
  Compute empirical ACF and PACF
  Propose candidate orders (p, q) with p ≤ p_max, q ≤ q_max
STEP 2 — Estimation
  For each candidate (p, q):
    Estimate parameters by maximum likelihood
    Compute AIC and BIC
STEP 3 — Diagnostics
  Compute residuals: ε̂_t = X_t − X̂_t
  IF autocorrelation detected in residuals:
    Return to STEP 1
STEP 4 — Forecasting
  Select model with lowest AIC or BIC
  Generate forecasts: X̂_{t+h|t} = E[X_{t+h} | F_t]
  Compute forecast intervals for each horizon h
OUTPUT: Validated ARMA(p*, q*) model and forecasts X̂_{t+h|t}
POSTCONDITION: Residuals with no significant autocorrelation
```



Non-Stationarity and Unit Root

When the AR model has a unit root, the process becomes non-stationary. The limit case is the random walk:

$$X_t = X_{t-1} + \varepsilon_t$$

Stationary Process

Model $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$

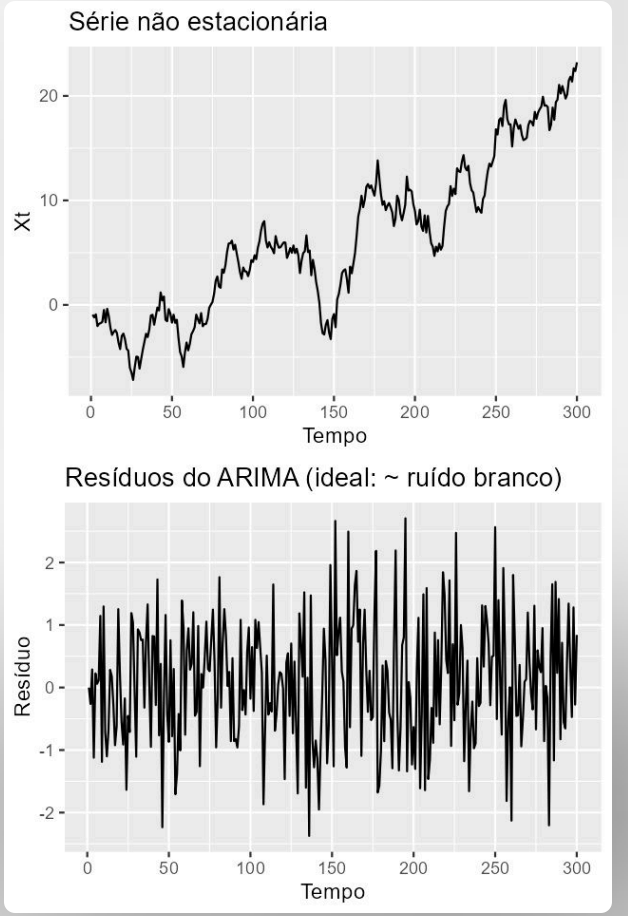
Unit Root

Limit case when $\phi = 1$, generating non-stationarity

Order of Integration

Number d of differencing operations required: $\Delta^d X_t$ is stationary

Non-stationarity arises when shocks have a permanent effect on the level of the series. The order of integration d measures how many differencing operations are needed to restore stationarity.



ARIMA Model

The ARIMA model integrates differencing and ARMA structure, representing the natural extension for non-stationary series:

$$\phi(B)(1-B)^d X_t = c + \theta(B)\varepsilon_t$$

AR Order (p)

Captures dependence on past values of the differenced series

Integration Order (d)

Number of differenciations to achieve stationarity

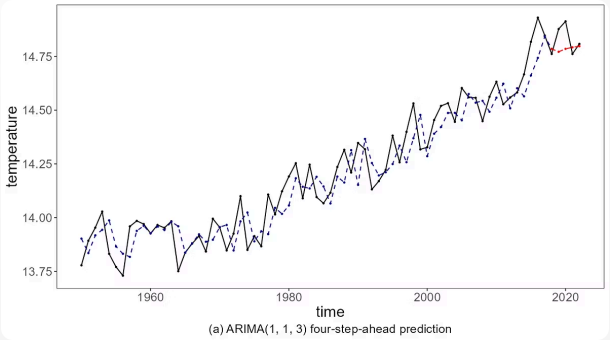
MA Order (q)

Models the propagation of random shocks

ARIMA Applications

ARIMA defines the expected behavior of the series, serving as a reference model. Events, anomalies, and structural changes may appear as systematic deviations from the pattern predicted by the model.

Linear models form the core of classical time series theory, providing the conceptual and methodological foundation for forecasting, structural analysis, and event detection in temporal data.



(a) ARIMA(1, 1, 3) four-step-ahead prediction

The Mathematics of Forecasting

Forecasting is estimating the future

$$\hat{X}_{t+h|t} = \mathbb{E}[X_{t+h} \mid \mathcal{F}_t]$$

We use all available information up to the present moment.

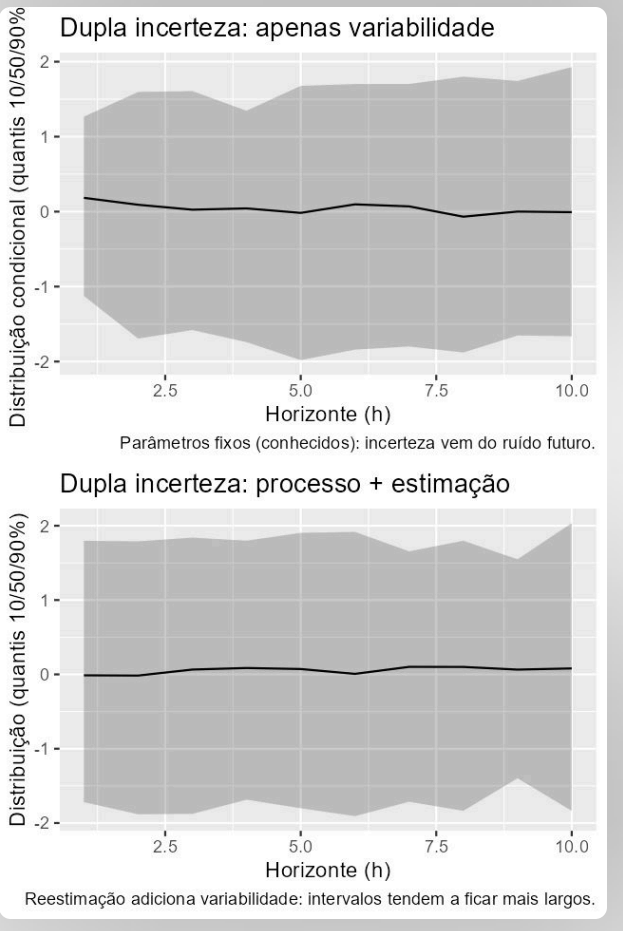
The model transforms the past into a forecast

The model's structure defines how historical data translates into future estimates.

The error measures the difference between predicted and observed

$$e_{t+h} = X_{t+h} - \hat{X}_{t+h|t}$$

Evaluating the error is evaluating the quality of the forecast.



Forecasting as an Inference Problem

Forecasting depends on a model fitted to the observed data. This model has parameters that need to be estimated, creating two sources of uncertainty: the natural variability of the process and the uncertainty in parameter estimation.

Forecast Function

The forecast is a function of the observed data and the estimated parameters

Estimated Parameters

We use $\hat{\theta}$ to represent the estimates of the model parameters

Dual Uncertainty

Uncertainty comes from the random behavior of the series and from parameter estimation

$$\hat{X}_{t+h|t} = g(X_t, X_{t-1}, \ldots; \hat{\theta})$$

Forecasting for Multiple Horizons

AR(1) Model

$$X_t = \phi X_{t-1} + \varepsilon_t$$

1-step Forecast

$$\hat{X}_{t+1|t} = \phi X_t$$

h-step Forecast

$$\hat{X}_{t+h|t} = \phi^h X_t$$

The multi-step forecast is obtained by recursive substitution: previous forecasts feed into the following ones. Uncertainty accumulates at each step.

$$\hat{X}_{t+h|t} = \phi^h \cdot X_t$$

ϕ is the autoregressive parameter. When $|\phi| < 1$, the forecast $\hat{x}[k]$ converges to zero in centered series (without intercept). If the model includes an intercept μ, the forecast converges to μ.

```
INPUT: x (observed series), ϕ (AR parameter), h (horizon)
PRE-CONDITION: fitted AR(1) model, |ϕ| < 1

STEP 1 — Initialization
  x_prev ← x[T]   // last observed value
  σ² ← estimated variance of model residuals
STEP 2 — Recursive generation of forecasts
  FOR k from 1 to h:
    x̂[k] ← ϕ · x_prev
    x_prev ← x̂[k]   // forecast becomes input for the next step
STEP 3 — Accumulated variance calculation
  FOR k from 1 to h:
    Var[k] ← σ² · SUM(ϕ^(2j), j=0..k-1)

OUTPUT: x̂[1..h] (forecasts), Var[1..h] (variances)
POST-CONDITION: x̂[k] → 0 as k → ∞ (series centered at zero);
                Var[k] grows monotonically with k.
```

Forecast as Conditional Distribution

A forecast is not just a number. The future is uncertain, and that uncertainty can be quantified.

Point Forecast

A single value summarizing the distribution, usually the conditional mean or median.

$$\hat{X}_{t+h|t} = E[X_{t+h} \mid \mathcal{F}_t]$$

Prediction Interval

A range of plausible values with a given probability of containing the true value.

$$\hat{X}_{t+h|t} \pm z_{\alpha/2}\sqrt{\mathrm{Var}(e_{t+h})}$$

Where $z_{\alpha/2}$ is the quantile of the normal distribution and α is the significance level.

Uncertainty

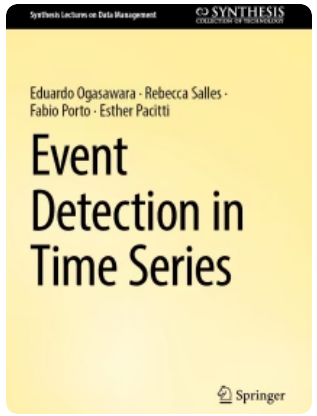
The variance of the forecast error, which grows with horizon *h*.

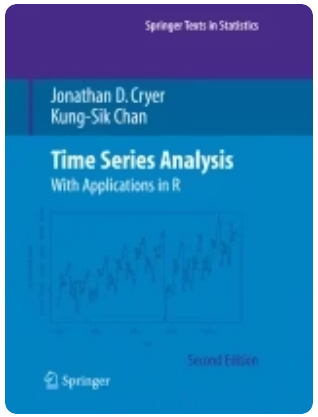
$$\mathrm{Var}(e_{t+h}) \text{ grows with } h$$

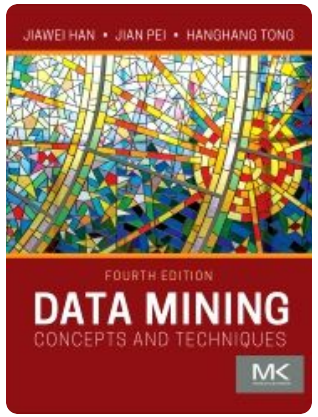
The farther the horizon, the wider the uncertainty band. Reporting only the point forecast hides essential information about forecast risk.

Fundamental property: the conditional expectation of the forecast error is zero — $\mathbb{E}[e_{t+h} \mid \mathcal{F}_t] = 0$ — but the variance grows with horizon *h*, reflecting increasing uncertainty in the distant future.

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